MIL-C-46594B (MU)
13 March 1969
SUPERSEDING
MIL-C-46594A (MU)
8 June 1967

MILITARY SPECIFICATION

CHARGE, DEMOLITION, SHAPED, 40 LB., M3A1 AND M3 PRACTICE, LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of demolition item designated as Charge, Demolition, Shaped, 40 Lb., M3A1 and M3 Practice.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition and Special Weapons; General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

FSC: 1375

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DRAWINGS

U. S. ARMY MUNITIONS COMMAND

7549165 - Box, Packing, Ammunition, for Charge Demolition, Shaped, 40 Lb., M3A1
7549166 - Support, Packing, Ammunition, for Charge Shaped, 40 Lb., M3A1
8858390 - Charge, Demolition, Shaped, 40 Lb., M3A1
9229253 - Charge, Demolition, Shaped, 40 Lb., M3 Practice
9230473 - Box, Packing, Ammunition for Charge, Shaped, 40 Lb., M3A1 (INERT)

PUBLICATIONS

U. S. ARMY MUNITIONS COMMAND

ET - 8858390 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material.- Materials shall be in accordance with applicable drawings and specifications.

3.2 Assemblies.- The assemblies shall comply with all requirements specified on Drawing (dwg.) 8858390 and with all requirements specified in applicable specifications.

3.3 Functioning.- The loaded assembly shall function satisfactorily as evidenced by complete burning of the charge and penetration of a 4 inch steel plate. The hole in the plate shall exhibit no evidence of double or multi-jets. This test shall be performed as specified in 4.2.3.3.

3.4 Workmanship.- All parts and assemblies shall be fabricated and loaded in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part

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nor shall the parts be contaminated by the cleaning agent. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.- At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.

b. Drawing and specification number and date, together with identification and date of changes thereto.

c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.

d. Number of items in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his

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principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection provisions

4.2.1 Lot formation.- The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot of loaded charge assemblies shall contain:

a. Parts from not more than one lot interfix number from one manufacturer.

b. Composition B of the same grade from one lot interfix number from one manufacturer which contains TNT from not more than one lot interfix number from one manufacturer.

c. Composition A-3 of the same grade from one lot interfix number from one manufacturer.

4.2.2 Examination.- The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major

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defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.2.4.

4.2.2.1 Booster (see dwg. 8858393 covering a detail of dwg. 8858390).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Concentricity of small cavity to outside diameter.....	Gage	01001
102.	Diameter of small cavity, minimum (min.).....	Gage	01002
103.	Depth of large cavity, min.....	Gage	01003
Minor:	AQL 1.50 percent		
201.	Depth of small cavity, min.....	Gage	01004
202.	Diameter of large cavity, min...	Gage	01005
203.	Total length, maximum (max.)....	Gage	01006

4.2.2.2 Assembly, prior to loading (see dwg. 8858390 or 9229253, as applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		

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Major:	AQL 0.40 percent		
101.	Interior coating with bare spot.	Visual	02001
102.	Foreign matter in interior.....	Visual	02002
Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4).....	Visual	02003

4.2.2.3 Assembly, prior to closing (see dwg. 8858390 or 9229253, as applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Charge improper.....	Visual	03001
102.	Booster missing (M3A1 only).....	Visual	03002
103.	Pad or spacer missing (M3A1 only).....	Visual	03003
104.	Container damaged.....	Visual	03004
Minor:	AQL 1.00 percent		
201.	Depth to charge.....	Gage	03005
202.	Evidence of poor workmanship (see 3.4).....	Visual	03006

4.2.2.4 Assembly (see dwg. 8858390 or 9229253, as applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Gage does not fully enter detonator well.....	Gage	04001
102.	Closing cap assembly not tight or adhesive missing.....	Visual-Manual	04002
103.	Protective tape missing.....	Visual	04003

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4).....	Visual	04004

4.2.2.5 Loading and marking assembly (see dwg. 8858390 or 9229253, as applicable).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Part number improper.....	Visual	05001
102.	Color coding improper.....	Visual	05002
103.	Component missing.....	Visual	05003
104.	Strap-slide or stand-off frame improperly assembled.....	Visual	05004
Minor:	AQL 1.50 percent		
201.	Total length, max.....	Gage	05005
202.	Width at stand-off legs, max....	Gage	05006
203.	Marking missing, misleading or unidentifiable.....	Visual	05007
204.	Evidence of poor workmanship (see 3.4).....	Visual	05008

4.2.2.6 Box, packing, prior to closing (see dwgs. 7549165 and 7549166).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Support missing.....	Visual	06001
102.	Packing component missing.....	Visual-Manual	06002
Minor:	None defined		

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4.2.2.7 Box, packing, sealed (see dwg. 7549165 and dwg. 9230473).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 1.00 percent		
101.	Contents of box exposed.....	Visual	07001
102.	Board broken.....	Visual	07002
103.	FSC number missing or misleading.....	Visual	07003
104.	Hardware missing, broken or loose.....	Visual-Manual	07004
105.	Strapping missing, broken or loose.....	Visual-Manual	07005
106.	Contents loose when box is inverted.....	Manual	07006
Minor:	AQL 2.50 percent		
201.	Marking missing, misleading or unidentifiable.....	Visual	07007
202.	Hardware improperly engaged.....	Visual-Manual	07008
203.	Strapping improperly engaged....	Visual-Manual	07009
204.	Car seal missing, unsealed or improperly positioned.....	Visual-Manual	07010
205.	Handle missing or insecure.....	Visual-Manual	07011

4.2.3 Testing

4.2.3.1 Specific gravity of the booster (see dwg. 8858393) - Major defect - Code No. 01007.- A sample of 5 boosters shall be selected from each lot and tested as specified in 4.3.1 using the applicable equipment of 4.2.4. If one or more units of the sample fail to meet the requirements, the lot shall be rejected.

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4.2.3.2 Weight of charge (see dwg. 8858390) - Major defect - Code No. 04006.- A sample of 25 shaped charges shall be selected from each lot for this determination. (The average weight of 30 metal parts assemblies shall be subtracted from each individual weight in making the determination.) If one or more units of the sample fail to meet the minimum requirement, those units shall be classed defective and the lot rejected subject to 100 percent charge weight determination. The scale used in making the determination shall be accurate to one eighth pound.

4.2.3.3 Functioning (see 3.3) - Major defect - Code No. 04007.- A sample of five (5) loaded assemblies shall be selected from each lot beginning with the first lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified. Failure of one or more assemblies to meet the specified requirement shall be cause for rejection of the lot represented by the sample. After three (3) consecutive lots have met this criteria, a sample of 5 loaded assemblies shall be selected from each fifth lot produced and shall be function tested. If one or more loaded assemblies fail to function, the sampling shall return to 5 loaded assemblies from each lot produced and continuing until three (3) consecutive lots have complied with the acceptance criteria specified. Testing shall be as specified in 4.3.2.

4.2.4 Inspection equipment.- Equipment Tabulation Number ET - 8858390 identifies the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.2.4.1 Government rights to documentation.- Inspection equipment drawings and lists provided and revised in accordance with the requirements of the ET may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey and whenever inspectionequipment drawings are needed.

4.3 Test methods and procedures

4.3.1 Specific gravity of boosters.- The booster shall be weighed on a balance and the weight recorded as dry weight. It shall then be weighed completely submerged under water, the temperature of which shall be 68 degrees Fahrenheit (F.) plus or minus 2 degrees F., and the weight recorded as weight under water. (When submerged, the position of the booster shall be such that no air is entrapped.) Specific gravity shall be calculated as follows:

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$$\text{Specific gravity} = \frac{A}{A-B}$$

where:
A = dry weight.
B = weight under water.

4.3.2 Functioning.- The loaded assembly shall be assembled for firing and positioned on the steel plate. A M6 electric blasting cap shall be secured in the capwell and the unit initiated. Observation shall be made for the presence of unburned charge and for evidence of double or multi-jets. Presence of either shall class a unit defective.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A.- The M3A1 Loaded Assembly shall be packed in accordance with dwg. 7549165.

5.1.2 Level C - The M3 Practice Assembly shall be packed in accordance with dwg. 9230473.

5.2 Marking

5.2.1 The M3A1 Loaded Assembly shall be marked in accordance with instructions contained in dwg. 7549165.

5.2.2 The M3 Practice Assembly shall be marked in accordance with instructions contained in dwg. 9230473.

6. NOTES

6.1 Ordering data.- Procurement documents shall specify the following:
a. Title, number and date of this specification.
b. Data cards.- Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

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6.2 Inspection code numbers.- The five digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. In addition, they should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.5).

6.3 Inspection equipment.- The contractor shall design inspection equipment as required by the inspection Equipment Lists (EL) referenced on the applicable ET in accordance with the instructions of paragraphs 6.3.1 through 6.3.5.

6.3.1 Inspection equipment lists (EL).- Inspection equipment lists indicate the availability of inspection equipment designs by showing in the "number" column of the list of inspection equipment (Form SMUPA 1010) the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.2. Design action required of the contractor is described in paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the number column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.2 Inspection equipment list codes.- The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.- Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications; mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an inspection

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equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor prepared designs or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.5.

6.3.4 Contractor designs.- Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finish, notes, operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to MIL-D-1000, Category A, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high contrast reproductions. Contractors shall submit three copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.5.

6.3.5 Submission of contractor designs.- All submitted designs shall contain a reference to the applicable five digit Code Number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND15. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract; submission shall be as directed by the contracting officer. Partial submission of inspection equipment designs is

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permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally within one month after receipt.

Custodian:
Army-MU

Preparing activity:
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Project Number: 1375-A-038

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

[ ] DIRECT GOVERNMENT CONTRACT [ ] SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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